

Part E: Summary & Reflection

This project set out to turn a raw, 39 column dataset into a clean, feature-engineered matrix for predicting employee attrition, and the process surfaced a series of decisions typical of real-world feature engineering work. The most consequential early decisions concerned data quality: four columns carried moderate, five to seven percent missingness, and rather than assuming a fix, the pattern of missingness itself was tested against the target before choosing a remedy. Finding no systematic relationship, straightforward median imputation was used in place of more elaborate group-wise or model-based approaches, since the evidence did not justify the added complexity. Categorical variables were split between one-hot encoding for genuinely nominal fields and explicit ordinal encoding for fields with a real ordering, and for a borderline case such as `Business_Travel`, the choice was grounded in whether the data actually showed a monotonic relationship with attrition rather than assumed from the category names alone. Eleven skewed numeric columns were normalized with a $\log_{1p}$ transform, chosen for being simple, invertible, and well-behaved at zero for the several zero-inflated columns present in the data, such as stock option value.

The construction and selection of features is where the most genuine challenges emerged. Several new ratio-based features were engineered from raw values, including compensation relative to experience, promotion velocity, and a combined overtime and work-life-balance burnout score, each grounded in a specific HR hypothesis about attrition. One of these ratios initially produced extreme skew traced back to a handful of employees with zero recorded experience but very large stock grants, a reminder that engineered ratios can introduce their own instability even when every input looks reasonable individually. This was resolved with a targeted log transform rather than discarding the feature outright. A more unexpected challenge came from redundancy hidden inside the supplied data itself: two columns, `Annual_Compensation` and `Engagement_Index`, turned out to be exact mathematical functions of other columns already present, discovered only by noticing infinite variance inflation factors and then confirming the underlying formulas directly. Systematic, iterative VIF-based elimination, applied without exception to both original and newly engineered columns, then removed several further redundant features, including one of the newly engineered ratios itself, evidence that constructed features need to be judged on their own statistical merit rather than protected simply because they were purpose-built.

The resulting preprocessing pipeline offers practical benefits beyond a tidier dataset. Every step, from imputation through the final principal-component reduction of six correlated satisfaction scores, is implemented as a small, reusable transformer inside a single scikit-learn pipeline object, so that statistics such as medians, encoder categories, scaling parameters, and the PCA axis are learned only from training data and then applied

unchanged to new data, avoiding leakage. Because each stage accepts and returns a labelled DataFrame rather than an anonymous array, intermediate results stay interpretable and easy to audit, and the pipeline was validated by confirming it reproduces the manual, step-by-step feature engineering exactly and generalizes cleanly to an unseen train-test split.

A few assumptions and limitations are worth stating plainly. Missingness was treated as random based on the checks performed, a reasonable but not certain conclusion, a different pattern in future data could call for a different imputation strategy. The ordinal ordering assigned to Education_Level rests on conventional academic hierarchy rather than an empirical relationship found in this dataset, where education level notably did not track compensation. The variance inflation factor threshold of ten and the Kaiser criterion used to size the principal component analysis are standard, defensible conventions rather than uniquely correct choices, and since this assignment was explicitly not evaluated on predictive accuracy, none of these decisions were validated against an actual attrition model, which remains a natural next step once this pipeline is put into use.